# MM-DINO: DINOv3-Based Universal Framework for Uni and Multimodal Remote Sensing Image Semantic Segmentation

Yuan Qin, Chanling Pan, Jinyun Chen, Ruibo Chen, Jiaxing Chen, and Ruichao Qu

***Abstract*—Semantic segmentation of high-resolution remote sensing imagery faces core challenges of scarce annotated data and weak model generalization. Although leveraging large-scale pretrained foundation models is considered key to breaking through these bottlenecks, directly adapting them to remote sensing tasks still faces three major issues: architectural mismatch, modality rigidity, and the difficulty of balancing efficiency with generalization. To address these, this article proposes MM-DINO, a universal and efficient framework based on DINOv3 for uni-modal and multimodal remote sensing image semantic segmentation. The framework employs a "Frozen Backbone-Adapter-Decoder" design: first, the pretrained DINOv3 backbone is kept entirely frozen to preserve its general visual knowledge; second, a modality-adaptive adapter is designed to transform sequential features into spatial pyramid features and enable early, soft cross-modal interaction via learnable weights; finally, a feature enhancement and refinement decoder is responsible for multiscale context aggregation, adaptive multimodal fusion, and progressive feature refinement. Extensive experiments on the ISPRS Vaihingen, Potsdam, and WHU-OPT-SAR datasets demonstrate the effectiveness of MM-DINO. Under the uni-modal setting, our method achieves mIoUs of 83.93%, 86.49%, and 55.72% on the three datasets, respectively, while under the multimodal setting, it achieves 84.32%, 86.54%, and 55.92%, all outperforming current state-of-the-art methods. Most notably, in zero-shot cross-dataset generalization experiments (trained on Vaihingen and tested on Potsdam), our method achieves 35.80% mIoU, significantly surpassing existing approaches and demonstrating remarkable domain robustness. Furthermore, efficiency analysis indicates that the framework achieves a favorable balance between accuracy and computational cost. The code will be open at: https://github.com/KimotaQY/MM-DINO**

***Index Terms*—DINOv3, domain generalization, foundation models, multimodal fusion, parameter-efficient fine-tuning, remote sensing imagery, semantic segmentation.**

## I. Introduction

ACCURATE semantic segmentation of high-resolution remote sensing imagery is a fundamental technology for the automated interpretation of land-cover information, supporting applications such as urban planning, environmental monitoring, and disaster assessment [1]. However, the performance and reliability of this technology have long been constrained by two major challenges. First, the acquisition of large-scale, high-quality, and finely annotated remote sensing data is costly and labor-intensive [2], [3]. Second, remote sensing images exhibit significant geographical, seasonal, and sensor-specific variations, leading to a severe domain generalization problem for models trained on limited data, often resulting in substantial performance degradation in new scenarios (shown in Fig. 1).

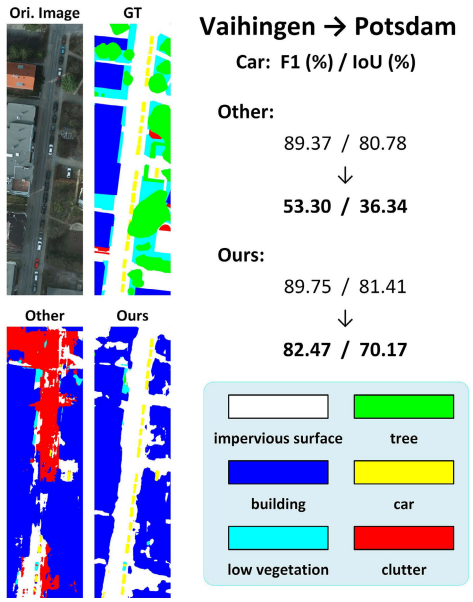



Fig. 1. Cross-dataset generalization results (trained on Vaihingen and tested on Potsdam). Visualized comparisons are provided to intuitively show the segmentation performance of uni-modal and multimodal methods on the Potsdam test set.

To alleviate the heavy reliance on annotated data, researchers are actively exploring the use of vision foundation models pretrained on massive datasets. These models, pretrained on vast datasets, learn general visual representations that are widely considered key to enhancing the performance and generalization of downstream tasks. Currently, the development of foundation models for remote sensing primarily follows two paths. The first involves training specialized remote sensing foundation models [4], [5], [6], [7], [8]. However, these are often hindered by nonpublic training

Received 14 December 2025; revised 10 March 2026; accepted 22 March 2026. Date of publication 25 March 2026; date of current version 10 April 2026. This work was by the Guangxi Key Technologies Research and Development Program under Grant AB23026015. *(Corresponding author: Chanling Pan.)*

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Digital Object Identifier 10.1109/TGRS.2026.3677346

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datasets or closed-source model weights, making them difficult for the broader academic community to widely validate, fairly compare against, and iteratively improve upon. The second path involves training on publicly available satellite image datasets. Yet, the extremely large computational resources and high engineering complexity required for this process present a nearly insurmountable barrier to replication for most research institutions. Consequently, adapting fully public, architecturally mature, and extensively validated general-purpose vision foundation models [9], [10], [11], [12] to remote sensing tasks has emerged as a more feasible and community-valuable pragmatic choice. Among these, the DINOv3 [13] model has released publicly available weights fine-tuned on satellite data, demonstrating exceptional potential for remote sensing feature extraction and providing an ideal starting point for building high-performance, strongly generalizable downstream applications.

Despite their promising prospects, directly applying these foundation models—originally designed for natural images—to remote sensing semantic segmentation presents three core challenges as follows.

*1) Architectural and Representational Gap:* Vision transformer backbones like DINOv3 are optimized for image-level understanding and output sequential patch embeddings. However, remote sensing semantic segmentation demands high-resolution, dense per-pixel predictions that preserve fine-grained spatial details and complex scene structures (e.g., building boundaries, road networks, and small objects like vehicles). Directly utilizing these sequential features can lead to the loss of critical spatial details and contextual relationships. This architectural mismatch between the backbone's representation format and the task's output requirements constitutes a fundamental obstacle.

*2) Modality Rigidity and Inflexible Fusion:* Remote sensing uniquely benefits from the synergistic fusion of multisource data, such as optical imagery (RGB/multispectral) and digital surface models (DSMs), providing complementary spectral and geometric information. However, most existing adaptation paradigms for foundation models are designed exclusively for single-modality inputs [14], [15]. While some specialized segmentation networks support multimodal fusion [16], [17], [18], their fusion mechanisms are often deeply coupled with specific backbone architectures and require end-to-end training. This modality rigidity makes it nontrivial to extend them to a frozen foundation model setting, let alone to flexibly accommodate an arbitrary number of input modalities. The absence of a lightweight, plug-and-play fusion mechanism compatible with frozen backbones severely limits the upper bound of model performance in real-world scenarios where diverse data sources are available.

*3) Trilemma of Efficiency, Generalization, and Adaptation:* Leveraging large foundation models involves navigating a delicate tradeoff. Full-parameter fine-tuning, while often delivering strong task-specific performance, is computationally prohibitive and risks catastrophic forgetting, eroding the very general visual knowledge that makes these models valuable. Conversely, simple linear probing fails to capture the complex spatial and contextual patterns required for dense prediction tasks, resulting in limited performance. The core question is: how can we introduce a minimal set of trainable parameters to effectively adapt the model to remote sensing segmentation, while maximally preserving and activating the foundation model's inherent, powerful generalization capability? Striking this balance is the key to developing practical, deployable solutions.

To address the aforementioned problems, this article proposes a universal framework named MM-DINO, designed to provide a powerful, efficient, and scalable solution for DINOv3-based remote sensing image semantic segmentation. Our core design philosophy is to keep the pretrained DINOv3 backbone entirely frozen to fully preserve its general visual knowledge, while designing a lightweight, task-specific, and adaptive head dedicated to handling dense prediction and multimodal fusion. Specifically, the contributions of this article are summarized as follows.

*1) Unified Frozen Backbone-Adapter-Decoder Architecture:* We present, for the first time, a complete decoding solution tailored for DINOv3 that keeps the backbone entirely frozen while natively supporting both uni-modal and arbitrarily many multimodal inputs. This decoupled design preserves the model's general visual knowledge while enabling task-specific adaptation.

*2) Modality-Adaptive Adapter With Early Cross-Modal Interaction:* We design a lightweight adapter that transforms sequential transformer outputs into spatial pyramid features—bridging the fundamental representational gap—and enables soft, learnable cross-modal interaction at an early stage, all without modifying the frozen backbone.

*3) Parameter-Efficient Enhancement and Refinement Decoder:* We construct a decoder comprising three synergistic modules—feature reinforcement module (FRM, cross-scale context aggregation), adaptive channel fusion module (ACFM, adaptive channel-wise fusion), and feature refinement neck (FRN, progressive feature refinement)—that achieves deep multimodal fusion and spatial refinement with only 31-M trainable parameters, maximally preserving the backbone's inherent generalization capability.

*4) State-of-the-Art Performance and Superior Generalization:* Extensive experiments on ISPRS Vaihingen, Potsdam, and WHU-OPT-SAR demonstrate that MM-DINO achieves state-of-the-art results in both uni-modal and multimodal settings. Notably, in zero-shot cross-dataset generalization, our method exhibits remarkable domain robustness, significantly outperforming existing approaches while maintaining a favorable accuracy-efficiency tradeoff.

The remainder of this article is organized as follows. Section II systematically reviews related work. Section III details the proposed MM-DINO framework. Section IV presents and analyzes the experimental results in depth and outlines potential directions for future research. Section V concludes the article.

## II. Related Work

### A. Remote Sensing Image Semantic Segmentation

The field has evolved from encoder–decoder architectures, typified by U-Net [19], to hybrid models incorporating attention mechanisms [20] and transformers [21], [22],

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[23], [24], with recent explorations into state-space models [25], [26], [27]. While these methods continue to achieve higher accuracy on specific benchmarks, their parameters are typically initialized from scratch or via pretraining on general-purpose datasets like ImageNet. This heavily supervised, domain-specific training paradigm results in learned feature representations with limited generalization capability, making them less effective when applied across different geographical regions or sensor types.

### B. Vision Task Adaptation Using Foundation Models

To overcome data limitations, leveraging large-scale pretrained foundation models has become a prominent research direction. Related work falls into two main categories as follows.

*1) General-Purpose Models and Their Adaptation:* General vision foundation models, such as SAM [9] and DINOv3 [13], are adapted to downstream tasks via prompt learning [28], [29] or parameter-efficient fine-tuning strategies [30], [31], [32]. For instance, SegDINO [33] demonstrated the potential of DINOv3 as a universal feature extractor, but its decoder design remains preliminary and does not fully account for the characteristics and multimodal nature of remote sensing data.

*2) Limitations of Remote Sensing-Specific Models:* Recent advances in remote sensing foundation models, such as SpectralGPT [34], have demonstrated the power of pretraining specifically on large-scale spectral data, achieving strong performance on spectral-dominated tasks. However, such specialized models are inherently tailored to the spectral domain and may not readily generalize to other modalities. Meanwhile, many other attempts to build remote sensing-specific foundation models still face challenges such as nonpublic training data and weights, or prohibitively high training costs that hinder reproducibility. This limits their broad accessibility and potential for deep innovation within the research community. Consequently, developing efficient adaptation solutions based on fully public, high-performance general foundation models like DINOv3—which offer broadly learned visual representations that can be efficiently transferred to multimodal remote sensing inputs via lightweight adapters—holds more immediate practical significance and community value.

### C. Multimodal Remote Sensing Data Fusion

Effectively fusing multisource information is crucial for improving segmentation robustness. Mainstream methods focus on feature-level fusion, implemented through designed two-stream networks [16], gating mechanisms [17], or transformer-based interaction modules [18]. Recently, the community has witnessed a surge in more sophisticated fusion paradigms to address specific challenges in complex environments. For instance, to tackle the issue of fragmented and small-scale objects in hilly agricultural regions, CM-Net [35] was proposed. It introduces a diverse receptive fusion module to adapt to varying object scales in optical images, coupled with a distributed weight attention module to enhance features of small targets. Similarly, to enhance cross-domain generalization across different cities, frameworks like HighDAN [36] employ adversarial learning to align multimodal feature representations. Concurrently, in the hyperspectral domain, S2DBFT [37] advances fusion by constructing a dual-branch transformer with a multihead spectral–spatial self-attention mechanism. However, existing approaches primarily rely on customized networks trained end-to-end. Their fusion mechanisms lack extensibility designs tailored for foundation model architectures and fail to systematically investigate how to achieve efficient and adaptive cross-modal complementarity after integrating foundation models.

In summary, a clear research gap remains: no lightweight yet powerful decoder exists for publicly available foundation models like DINOv3 that can simultaneously address the three core challenges: 1) transforming sequential features for dense prediction; 2) enabling flexible and extensible adaptive multimodal fusion; and 3) maximizing the preservation of the model's inherent generalization capability with minimal trainable parameters. This work aims to systematically address this gap.

## III. Methodology

This chapter details the proposed MM-DINO framework. As illustrated in Fig. 2, the overall architecture consists of three core components: 1) a frozen pretrained vision transformer backbone for extracting powerful general-purpose visual features; 2) a modality-adaptive adapter responsible for projecting features from different modalities into a unified spatial scale and facilitating preliminary cross-modal interaction; and 3) a feature enhancement and refinement decoder, which performs deep fusion of multiscale, multimodal features followed by semantic refinement to generate the final segmentation map.

### A. Overall Architecture Overview

Given an input image $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, MM-DINO supports both single-modal ($C = 3$) and multimodal scenarios. In single-modal mode, only the primary modality path is used. In multimodal mode, all modalities are first independently fed into a shared, weight-frozen DINOv3 backbone to extract multilevel features. These features are then passed to the modality-adaptive adapter. This module not only reconstructs the sequential features into spatial feature maps and aligns their channels but also enables soft information exchange between modalities via learnable adaptive weights. The processed multiscale features are subsequently input to the feature enhancement and refinement decoder. This decoder first enriches the contextual information of each scale via the FRM, then integrates information from different modalities using the improved ACFM, and finally performs progressive feature fusion and upsampling through the FRN to produce the final segmentation prediction.

### B. Frozen Backbone and Feature Extraction

To leverage the generic visual knowledge embedded in large-scale pretrained models while avoiding catastrophic forgetting, we employ DINOv3-ViT-L/16 [13] as the backbone and keep its weights entirely frozen during downstream task training. Given an input modality $\mathbf{X}_m$, we feed it into the backbone and extract the output features from four specific intermediate transformer blocks, denoted as $\{\mathbf{F}_m^i\}_{i=1}^4$, where

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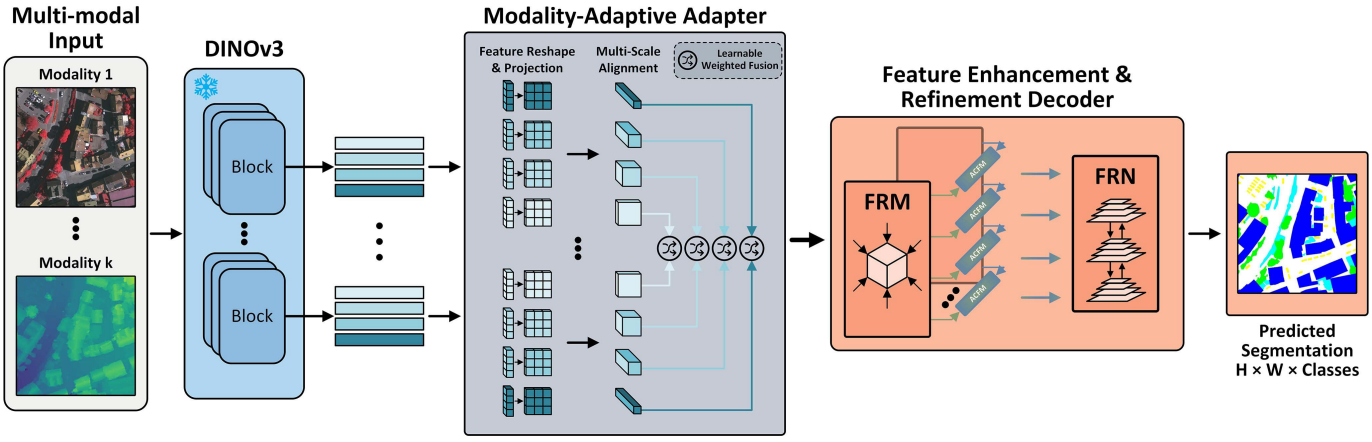



Fig. 2. Overview of the proposed MM-DINO framework for uni-modal and multimodal remote sensing image semantic segmentation. The architecture consists of three main components: 1) a frozen pretrained DINOv3 backbone that extracts multilevel visual features from input modalities; 2) a modality-adaptive adapter that reshapes, aligns, and performs early cross-modal interaction through learnable adaptive weights; and 3) a feature enhancement and refinement decoder composed of an FRM, an ACFM, and an FRN for multiscale fusion and upsampling to produce the final segmentation map. The framework supports both single-modal (e.g., RGB only) and multimodal (e.g., RGB + DSM) inputs, as illustrated by the separate modality paths that converge at the adapter stage. Arrows indicate the flow of features through the network.

$\mathbf{F}_m^i \in \mathbb{R}^{B\times L\times D}$, $B$ is the batch size, $L = (H/16) \times (W/16)$ is the sequence length, and $D = 1024$ is the feature dimension. These features capture rich visual information ranging from low-level details to high-level semantics.

### C. Modality-Adaptive Adapter

This module aims to transform the sequential features from the backbone into spatial multiscale pyramid features suitable for dense prediction tasks and provide a flexible interaction mechanism for multimodal inputs.

*1) Feature Reshaping and Projection:* For the $i$th layer feature $\mathbf{F}_m^i$ of each modality, we first reshape it into a 2-D spatial feature map $\mathbf{C}_m^i \in \mathbb{R}^{B\times D\times H_i\times W_i}$, where $H_i = H/16, W_i = W/16$. Subsequently, a lightweight $1\times 1$ convolutional layer $\text{Proj}^i$ projects it to a predefined channel dimension $C_i$, forming the initial multiscale features $\{\mathbf{P}_m^i\}_{i=1}^4$.

*2) Multiscale Resolution Alignment:* To construct a feature pyramid with consistent semantic granularity, we design a set of upsampling or downsampling operations $\text{Resize}^i$ (including transposed convolution, standard convolution, or identity mapping) to adjust $\{\mathbf{P}_m^i\}$ to a common intermediate scale, outputting the aligned features $\{\widetilde{\mathbf{P}}_m^i\}_{i=1}^4$.

*3) Multimodal Adaptive Interaction:* In multimodal scenarios, to achieve complementary information exchange at an early fusion stage, we introduce a lightweight adaptive fusion mechanism. Assuming $N$ input modalities, for the $i$th scale, we assign a learnable scalar weight $\alpha_k^i$ (initialized as $1/N$) to each modality $k$, with $\sum_{k=1}^N \alpha_k^i = 1$. The interacted feature $\widehat{\mathbf{P}}_k^i$ for modality $k$ is obtained via a weighted sum of all modalities' features at that scale

$$\widehat{\mathbf{P}}_k^i = \sum_{j=1}^{N} \alpha_{kj}^i \cdot \widetilde{\mathbf{P}}_j^i. \tag{1}$$

Here, the weight $\alpha_{kj}^i$ denotes the contribution of the $j$th modality's feature when fusing into the $k$th modality's feature, obtained by applying a softmax layer over all $j$. This allows each modality to adaptively absorb complementary information from others, resulting in an enhanced set of multiscale features $\widehat{\mathbf{P}}_k^i$ with preliminary interaction for the decoder.

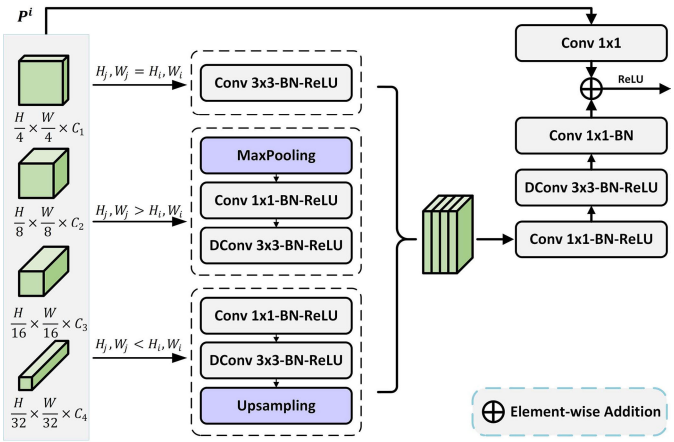



Fig. 3. Structure of the FRM. This module enriches the contextual representation of each input feature scale by generating multireceptive-field contextual features through parallel convolutional branches. The aligned outputs from all branches are concatenated and then fused with the original inputs via a lightweight residual fusion block, producing enhanced features that preserve fine details while integrating rich context.

### D. Feature Enhancement and Refinement Decoder

The decoder takes the multiscale features from the adapter and progressively performs enhancement, fusion, and refinement.

*1) Feature Reinforcement Module:* Inspired by [38], we design the FRM (shown in Fig. 3) to explicitly aggregate contextual information from different scales, thereby enhancing the representational power of each individual scale feature. For the input $i$-th scale feature $\widehat{\mathbf{P}}_k^i$, the FRM generates multiresolution features through a set of parallel operations with varying strides (pooling or convolution). These features

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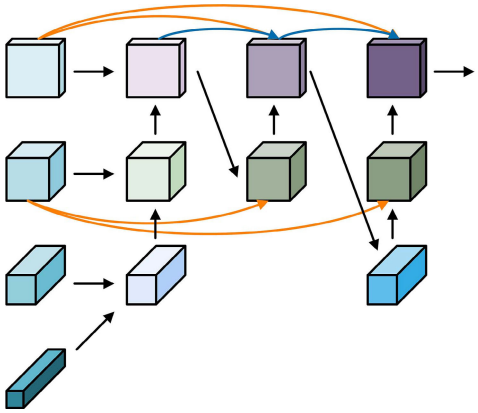


Fig. 4. Structure of the FRN. This module adopts a strategy combining top-down and iterative refinement. First, high-level semantic features are upsampled and fused with lower level detailed features step by step (top-down pathway). Then, local iterative units are introduced at key levels, progressively enhancing the semantic consistency and boundary accuracy of features by repeatedly fusing the current features, the refined output from the previous step, and the original backbone features.

are then aligned to the same resolution via upsampling or downsampling, concatenated, and finally processed by a fusion convolutional layer to produce the enhanced feature $S^i$. This process ensures that $S^i$ contains both fine-grained details from its own scale and contextual information from other scales.

*2) Multimodal Feature Fusion:* After obtaining the enhanced features $\{\mathbf{S}_m^i\}$ for each modality, we need to fuse them effectively. To this end, we design an ACFM that supports an arbitrary number ($N$) of modalities. For the $i$th scale, each modality's feature $S_m^i$ is first calibrated by an independent channel attention block (squeeze-and-excitation block [18]), which learns importance weights for each channel. The calibrated features from all modalities are then fused via element-wise summation

$$\mathbf{F}_{\text{fused}}^i = \sum_{m=1}^{N} \text{SE}_m\left(\mathbf{S}_m^i\right). \tag{2}$$

*3) Feature Refinement Neck:* Drawing inspiration from the feature pyramid refinement ideas in [39], we propose a lightweight FRN (shown in Fig. 4) to further improve the semantic consistency and spatial accuracy of the fused feature pyramid. The FRN adopts a progressive refinement strategy, which can be summarized in three key steps as follows.

*a) Top-down enhancement:* Starting from the highest semantic layer, features are upsampled and fused with features from the next layer, propagating rich contextual information downward.

*b) Cross-scale iterative refinement:* An iterative refinement structure is introduced at intermediate layers. It repeatedly fuses the current layer's feature, the detailed feature from the previous refinement step, and the original backbone feature to iteratively optimize the feature representation. This step is formulated as

$$\mathbf{P}_l^{t+1} = \mathcal{H}\left(\left[\text{Up}\left(\mathbf{P}_{l-1}^t\right), \mathbf{P}_l^t, \mathbf{P}_l^{\text{in}}\right]\right) \tag{3}$$

where $\mathcal{H}$ denotes a convolutional layer, $t$ is the refinement step index, and $\mathbf{P}_l^{in}$ is the original feature output by the FRM.

## IV. Experiments

### A. Data Description

Our experiments are conducted on three public and widely used remote sensing semantic segmentation datasets: ISPRS Vaihingen, Potsdam, and WHU-OPT-SAR. Their details are as follows.

*1) Vaihingen:* This dataset contains 16 high-resolution orthophotos with an average size of 2500 × 2000 pixels. Each image consists of three channels: near-infrared, red, and green (NIR-R-G), along with a corresponding normalized DSM (nDSM) at a 9-cm ground sampling distance. Following the standard split in literature [40], we use 12 images for training and the remaining 4 for testing.

*2) Potsdam:* This larger dataset consists of 24 orthophotos, each 6000 × 6000 pixels in size. The data includes four multispectral bands: infrared, red, green, and blue (IR-R-G-B), as well as an nDSM with a 5-cm resolution. To diversify our experiments, we use only the latter three bands (RGB) to simulate common visible light input. Similarly, 18 images are used for training and 6 for testing.

*3) WHU-OPT-SAR:* This dataset contains 100 optical images and SAR images, each with a size of 5556 × 3704 pixels and a ground sampling distance of 5 m. The optical imagery includes both RGB and near-infrared (NIR) channels. We use only the latter three bands (RGB) to simulate common visible light input. Following the standard protocol in [41], we use 80 images for training and the remaining 20 for testing.

### B. Implementation Details

All experiments are conducted in the same software and hardware environment to ensure fair comparison. Specific settings are as follows.

*1) Evaluation Metrics:* We employ the commonly used mean $F1$-score and mean intersection over union as the primary evaluation metrics, denoted as mF1 and mIoU, respectively, to comprehensively measure the model's classification accuracy and region overlap across all categories.

*2) Hardware and Software:* Experiments are performed on two NVIDIA L20 GPUs with 48-GB memory each, using the PyTorch framework.

*3) Training Strategy:* Input images are cropped into 512 × 512 pixel patches. We use the AdamW optimizer with an initial learning rate of $1e^{-4}$ and a weight decay of 0.01. The learning rate schedule follows a cosine annealing strategy. Models are trained for 50 epochs with a batch size of 8. To alleviate class imbalance, a joint loss function combining softmax cross-entropy loss and dice loss is used.

*4) Data Augmentation:* During training, random resizing, cropping, horizontal flipping, and vertical flipping are adopted for data augmentation. All compared methods are retrained or evaluated under the same data augmentation strategy, while their hyperparameters remain consistent with their original papers.

### C. Performance Comparison on Standard Datasets

We compare MM-DINO with state-of-the-art uni-modal (RGB only) and multimodal (RGB+DSM) remote sensing

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TABLE I

QUANTITATIVE RESULTS ON THE VAIHINGEN TEST SET. "A/B" REPRESENTS $F1$-SCORE/IOU FOR EACH CATEGORY. THE BEST SCORE IS HIGHLIGHTED IN BOLD, AND THE SECOND-BEST SCORE IS UNDERLINED

| Modality | Models | Backbone | impervious surface | building | low vegetation | tree | car | mF1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|
| Uni | UNetFormer$_{22}$ | ResNet18 | 90.06 / 81.92 | 94.40 / 89.39 | 77.26 / 62.95 | 89.28 / 80.64 | 77.66 / 63.48 | 85.73 | 75.67 |
| | CMTFNet$_{23}$ | ResNet50 | 93.37 / 87.57 | 96.84 / 93.88 | 81.28 / 68.46 | 91.18 / 83.79 | 89.37 / 80.78 | 90.40 | 82.89 |
| | RS3Mamba$_{24}$ | R18-Mamba-T | 91.58 / 84.47 | 95.75 / 91.84 | 78.66 / 64.83 | 89.75 / 81.41 | 78.45 / 64.54 | 86.84 | 77.42 |
| | UMFormer$_{25}$ | ResNet18 | 91.49 / 84.31 | 95.79 / 91.91 | 78.58 / 64.72 | 90.40 / 82.48 | 83.36 / 71.47 | 87.92 | 78.98 |
| | SegDINO$_{25}$ | DINOv3-L | 88.40 / 79.21 | 92.88 / 86.70 | 76.07 / 61.39 | 89.51 / 81.02 | 75.07 / 60.09 | 84.39 | 73.68 |
| | **Ours** | DINOv3-L | 93.37 / 87.57 | 97.41 / 94.95 | 81.03 / 68.11 | 91.24 / 83.89 | 89.75 / 81.41 | 90.56 | 83.19 |
| | **Ours** (LoRA) | | **93.75 / 88.24** | **97.56 / 95.24** | **81.74 / 69.13** | **91.42 / 84.20** | **90.61 / 82.83** | **91.02** | **83.93** |
| Multi | ESANet$_{21}$ | ResNet34 | 92.03 / 85.24 | 96.44 / 93.13 | 79.80 / 66.39 | 90.68 / 82.95 | 84.58 / 73.29 | 88.71 | 80.20 |
| | CMGFNet$_{22}$ | ResNet34 | 92.00 / 85.19 | 95.95 / 92.21 | 79.89 / 66.52 | 90.62 / 82.85 | 88.00 / 78.57 | 89.29 | 81.07 |
| | FTransUNet$_{24}$ | R50-ViT-B | 93.17 / 87.21 | 97.13 / 94.42 | 81.44 / 68.69 | 91.13 / 83.71 | 85.93 / 75.33 | 89.76 | 81.87 |
| | MultiSenseSeg$_{24}$ | Segformer-B2 | 92.22 / 85.57 | 96.17 / 92.63 | 80.28 / 67.06 | 90.76 / 83.08 | 80.20 / 66.95 | 87.93 | 79.06 |
| | MFNet$_{25}$ | SAM-L | 93.03 / 86.97 | 96.58 / 93.40 | **82.92 / 70.83** | **92.03 / 85.24** | 90.22 / 82.18 | 90.96 | 83.72 |
| | **Ours** | DINOv3-L | 93.10 / 87.09 | 97.15 / 94.45 | 81.59 / 68.90 | 91.48 / 84.29 | 90.88 / 83.28 | 90.84 | 83.60 |
| | **Ours** (LoRA) | | **93.87 / 88.44** | **97.68 / 95.47** | 81.97 / 69.45 | 91.46 / 84.26 | **91.28 / 83.96** | **91.25** | **84.32** |

TABLE II

QUANTITATIVE RESULTS ON THE POTSDAM TEST SET. "A/B" REPRESENTS $F1$-SCORE/IOU FOR EACH CATEGORY. THE BEST SCORE IS HIGHLIGHTED IN BOLD, AND THE SECOND-BEST SCORE IS UNDERLINED

| Modality | Models | Backbone | impervious surface | building | low vegetation | tree | car | mF1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|
| Uni | UNetFormer$_{22}$ | ResNet18 | 88.99 / 80.16 | 93.16 / 87.19 | 81.27 / 68.46 | 78.03 / 63.98 | 94.09 / 88.83 | 87.11 | 77.72 |
| | CMTFNet$_{23}$ | ResNet50 | **93.54 / 87.87** | **97.46 / 95.04** | 87.10 / 77.15 | 87.29 / 77.44 | **96.75 / 93.71** | 92.43 | 86.24 |
| | RS3Mamba$_{24}$ | R18-Mamba-T | 91.05 / 83.56 | 95.30 / 91.02 | 82.77 / 70.60 | 81.51 / 68.79 | 92.72 / 86.42 | 88.67 | 80.08 |
| | UMFormer$_{25}$ | ResNet18 | 90.04 / 81.89 | 94.74 / 90.01 | 83.20 / 71.23 | 81.79 / 69.20 | 94.25 / 89.13 | 88.81 | 80.29 |
| | SegDINO$_{25}$ | DINOv3-L | 86.59 / 76.35 | 89.29 / 80.65 | 80.43 / 67.26 | 81.26 / 68.44 | 90.18 / 82.11 | 85.55 | 74.96 |
| | **Ours** | DINOv3-L | 93.32 / 87.48 | 96.98 / 94.14 | 87.06 / 77.09 | 87.63 / 77.98 | 96.49 / 93.22 | 92.30 | 85.98 |
| | **Ours** (LoRA) | | 93.45 / 87.70 | 97.18 / 94.52 | **87.82 / 78.28** | **87.93 / 78.47** | 96.63 / 93.47 | **92.60** | **86.49** |
| Multi | ESANet$_{21}$ | ResNet34 | 91.33 / 84.04 | 96.03 / 92.36 | 84.95 / 73.84 | 85.76 / 75.07 | 95.00 / 90.47 | 90.61 | 83.16 |
| | CMGFNet$_{22}$ | ResNet34 | 92.35 / 85.79 | 96.76 / 93.72 | 85.59 / 74.81 | 85.98 / 75.40 | 94.80 / 90.11 | 91.10 | 83.97 |
| | FTransUNet$_{24}$ | R50-ViT-B | 91.10 / 83.66 | 94.51 / 89.60 | 84.58 / 73.27 | 85.53 / 74.71 | 93.33 / 87.50 | 89.81 | 81.75 |
| | MultiSenseSeg$_{24}$ | Segformer-B2 | 90.83 / 83.20 | 94.41 / 89.42 | 84.31 / 72.87 | 82.99 / 70.93 | 93.77 / 88.27 | 89.26 | 80.94 |
| | MFNet$_{25}$ | SAM-L | 92.85 / 86.66 | 97.09 / 94.35 | 86.89 / 76.82 | 87.64 / 78.00 | 96.22 / 92.71 | 92.14 | 85.71 |
| | **Ours** | DINOv3-L | 93.34 / 87.51 | 97.08 / 94.33 | 87.47 / 77.73 | 87.42 / 77.65 | **96.32 / 92.90** | 92.33 | 86.02 |
| | **Ours** (LoRA) | | **93.54 / 87.86** | **97.53 / 95.18** | **87.90 / 78.41** | **87.92 / 78.45** | 96.25 / 92.77 | **92.63** | **86.54** |

segmentation methods from recent years, including uni-modal methods (UNetFormer [42], CMTFNet [43], RS3Mamba [44], UMFormer [45], and SegDINO [33]), and multimodal methods (ESANet [16], CMGFNet [17], FTransUNet [18], MultiSenseSeg [46], and MFNet [40]). To fully assess the model's potential, we report two versions: Ours (training only the newly introduced modules with the backbone frozen) and Ours (LoRA) (applying LoRA fine-tuning to the attention layers on top of the frozen backbone, the rank $r$ of the low-rank matrices is set to 3). Results are summarized in Table I (Vaihingen), Table II (Potsdam), and Table III (WHU-OPT-SAR).

The methods compared in the article all use the officially released code and are trained under the same conditions.

*1) Quantitative Results: a) Analysis of Vaihingen results (Table I):* Under the uni-modal (Uni) setting, our model (Ours) achieves an mIoU of 83.19%, significantly outperforming other methods like CMTFNet (82.89%) and UMFormer (78.98%). Performance is further improved to 83.93% when LoRA fine-tuning is enabled, setting a new benchmark. Under the multimodal (Multi) setting, MM-DINO also demonstrates excellent performance. Our model achieves an mIoU of 83.60%, surpassing strong multimodal fusion networks such as FTransUNet (81.87%) and CMGFNet (81.07%). The Ours (LoRA) variant achieves the best performance among all methods (84.32% mIoU), even slightly outperforming MFNet (83.72%), which also utilizes a large pretrained model (SAM).

*b) Analysis of Potsdam results (Table II):* On the more challenging Potsdam dataset with larger scale variations, the advantages of MM-DINO become even more pronounced. In the uni-modal setting, our model (85.98% mIoU) performs comparably to the current best method, CMTFNet (86.24% mIoU), while Ours (LoRA) (86.49% mIoU) achieves superiority. In the multimodal setting, Ours (LoRA) achieves the best overall performance with 86.54% mIoU, attaining the highest $F1$-scores in several specific categories (e.g., impervious surface, building, and low vegetation).

*c) Analysis of WHU-OPT-SAR results (Table III):* This dataset exhibits distinct characteristics that set it apart from the ISPRS datasets: its optical imagery has a spatial resolution of only 5 m, which is substantially lower than the centimeter-level precision of Vaihingen and Potsdam. Under the uni-modal (optical RGB only) setting, our method (Ours) achieves

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TABLE III

QUANTITATIVE RESULTS ON THE WHU-OPT-SAR TEST SET. "A/B" REPRESENTS $F1$-SCORE/IOU FOR EACH CATEGORY. THE BEST SCORE IS HIGHLIGHTED IN BOLD, AND THE SECOND-BEST SCORE IS UNDERLINED

| Modality | Models | Backbone | farmland | city | village | water | forest | road | others | mF1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Uni | $\text{UNetFormer}_{22}$ | ResNet18 | 76.80 / 62.34 | 64.31 / 47.40 | 58.85 / 41.70 | 68.08 / 51.60 | 89.64 / 81.22 | 39.09 / 24.29 | 15.79 / 8.57 | 58.94 | 45.30 |
| | $\text{CMTFNet}_{23}$ | ResNet50 | 81.33 / 68.53 | 72.42 / 56.76 | 64.42 / 47.51 | 77.57 / 63.35 | 91.01 / 83.50 | 57.47 / 40.32 | 35.45 / 21.54 | 68.52 | 54.50 |
| | $\text{RS3Mamba}_{24}$ | R18-Mamba-T | 79.86 / 66.48 | 70.14 / 54.01 | 61.31 / 44.21 | 74.11 / 58.87 | 90.32 / 82.35 | 45.25 / 29.24 | 25.32 / 14.50 | 63.76 | 49.95 |
| | $\text{UMFormer}_{25}$ | ResNet18 | 78.56 / 64.69 | 68.86 / 52.51 | 61.32 / 44.21 | 71.85 / 56.06 | 90.10 / 81.98 | 49.35 / 33.29 | 24.24 / 13.79 | 63.55 | 49.51 |
| | $\text{SegDINO}_{25}$ | DINOv3-L | 75.92 / 61.19 | 67.75 / 51.22 | 56.44 / 39.31 | 65.35 / 48.53 | 89.27 / 80.62 | 34.47 / 20.83 | 8.98 / 4.70 | 56.88 | 43.77 |
| | **Ours** | DINOv3-L | 81.27 / 68.45 | 72.35 / 56.67 | 65.30 / 48.48 | 77.88 / 63.77 | 91.15 / 83.74 | 55.94 / 38.83 | **39.32 / 24.47** | 69.03 | 54.92 |
| | **Ours** (LoRA) | | **82.00 / 69.50** | **73.31 / 57.87** | **65.72 / 48.94** | **78.03 / 63.98** | **91.21 / 83.85** | **58.70 / 41.55** | 39.19 / 24.37 | **69.74** | **55.72** |
| Multi | $\text{ESANet}_{21}$ | ResNet34 | 79.33 / 65.75 | 70.40 / 54.32 | 61.66 / 44.57 | 73.87 / 58.57 | 90.48 / 82.62 | 42.98 / 27.37 | 25.55 / 14.64 | 63.47 | 49.69 |
| | $\text{CMGFNet}_{22}$ | ResNet34 | 80.21 / 66.96 | 68.52 / 52.12 | 63.00 / 45.99 | 75.46 / 60.59 | 90.72 / 83.02 | 51.42 / 34.61 | 30.84 / 18.23 | 65.74 | 51.65 |
| | $\text{FTransUNet}_{24}$ | R50-ViT-B | 79.99 / 66.66 | 69.58 / 53.35 | 62.15 / 45.09 | 74.31 / 59.13 | 90.38 / 82.44 | 35.75 / 21.76 | 0.06 / 0.03 | 58.89 | 46.92 |
| | $\text{MultiSenseSeg}_{24}$ | Segformer-B2 | 77.87 / 63.76 | 66.72 / 50.06 | 61.48 / 44.38 | 72.74 / 57.15 | 90.24 / 82.22 | 48.94 / 32.39 | 27.16 / 15.71 | 63.59 | 49.38 |
| | $\text{MFNet}_{25}$ | SAM-L | 81.40 / 68.63 | 72.02 / 56.28 | 63.87 / 46.92 | 76.32 / 61.71 | 90.86 / 83.26 | 55.14 / 38.06 | 29.69 / 17.43 | 67.04 | 53.18 |
| | **Ours** | DINOv3-L | 81.64 / 68.98 | 72.05 / 56.30 | 64.74 / 47.86 | 77.96 / 63.88 | 91.11 / 83.67 | 57.85 / 40.69 | 38.56 / 23.89 | 69.13 | 55.04 |
| | **Ours** (LoRA) | | **82.14 / 69.69** | **73.29 / 57.85** | **65.67 / 48.89** | **78.49 / 64.60** | **91.21 / 83.83** | **59.43 / 42.28** | **39.10 / 24.30** | **69.90** | **55.92** |

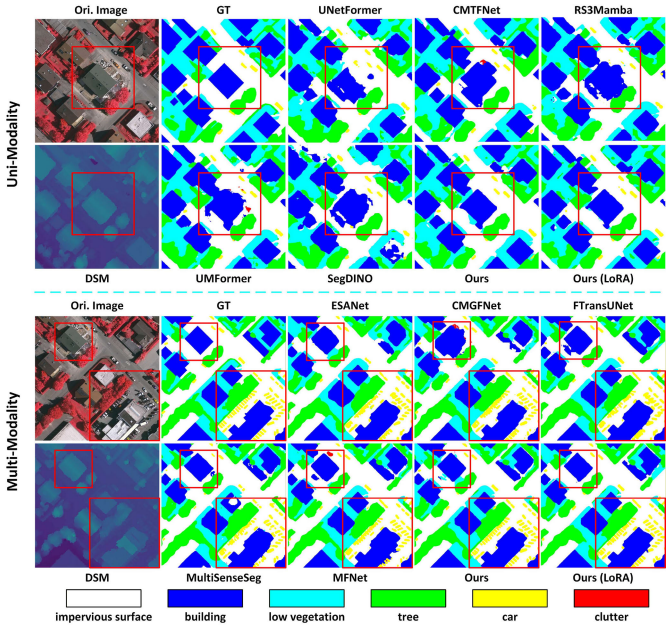



Fig. 5. Visualized comparisons on the Vaihingen test set. To highlight the differences, some red boxes are added to all subfigures. Specifically, the upper part of the figure presents the results of uni-modal methods, while the lower part shows those of multimodal methods.

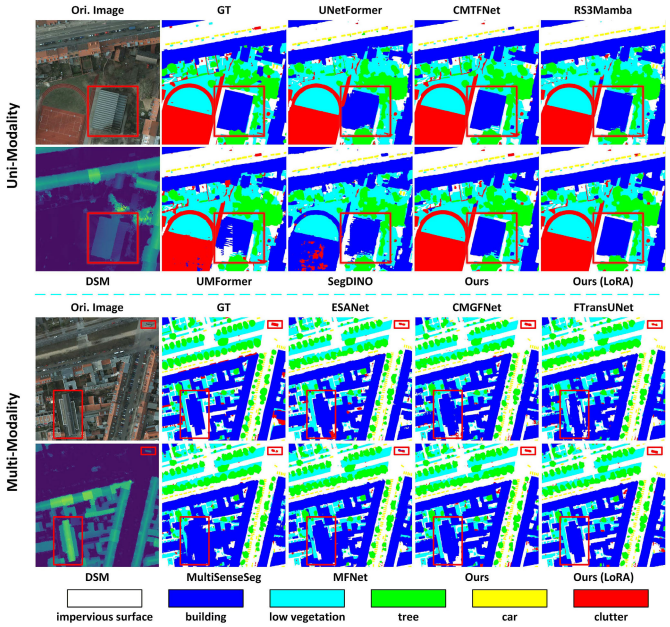



Fig. 6. Visualized comparisons on the Potsdam test set. To highlight the differences, some red boxes are added to all subfigures. Specifically, the upper part of the figure presents the results of uni-modal methods, while the lower part shows those of multimodal methods.

54.92% mIoU, significantly outperforming existing approaches such as CMTFTNet (54.50% mIoU). With the introduction of LoRA fine-tuning, Ours (LoRA) further improves performance to 55.72% mIoU, ranking first among all compared methods. In the multimodal (optical+SAR) setting, Ours (LoRA) leads with 55.92% mIoU, surpassing the second-best method MFNet (53.18% mIoU) by 2.74 percentage points. Notably, it demonstrates clear advantages on challenging categories such as "road" and "others," achieving $F1$-scores of 59.43% and 39.10%, respectively.

*2) Qualitative Results:* To provide intuitive visual evidence for the quantitative superiority of MM-DINO, we present representative segmentation results from the Vaihingen, Potsdam, and WHU-OPT-SAR datasets in Figs. 5–7, respectively.

Visually, MM-DINO generates predictions with superior geometric fidelity and semantic coherence. Specifically, it achieves more complete segmentation of large objects (e.g., buildings, impervious surfaces, and water) with fewer internal holes, sharper and more accurate boundaries that significantly reduce interclass adhesion, and consistently better detection and delineation of small, scattered targets such as vehicles. Furthermore, on the WHU-OPT-SAR dataset (see Fig. 7), our method demonstrates superior performance in extracting road networks, producing continuous and topologically correct roads with significantly fewer breaks compared to other methods, which further attests to its robustness across diverse modalities and resolutions.

### D. Evaluation of Cross-Dataset Generalization Capability

To evaluate the model's domain adaptation and generalization ability, we design a cross-dataset experiment: models are trained on the Vaihingen dataset and then directly evaluated on the Potsdam test set without any fine-tuning. This test

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TABLE IV

CROSS-DATASET GENERALIZATION RESULTS (TRAINED ON VAIHINGEN AND TESTED ON POTSDAM). "A/B" REPRESENTS $F1$-SCORE/IOU FOR EACH CATEGORY. THE BEST SCORE IS HIGHLIGHTED IN BOLD, AND THE SECOND-BEST SCORE IS UNDERLINED

| Modality | Models | Backbone | impervious surface | building | low vegetation | tree | car | mF1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|
| Uni | $UNetFormer_{22}$ | ResNet18 | 21.54 / 12.07 | 40.46 / 25.36 | 4.98 / 2.55 | 0.57 / 0.28 | 13.29 / 7.12 | 16.17 | 9.48 |
| | $CMTFNet_{23}$ | ResNet50 | 41.06 / 25.83 | 47.71 / 31.33 | 3.55 / 1.81 | 0.05 / 0.02 | 53.30 / 36.34 | 29.13 | 19.07 |
| | $RS3Mamba_{24}$ | R18-Mamba-T | 43.16 / 27.52 | 47.83 / 31.43 | 10.43 / 5.50 | 0.06 / 0.03 | 29.20 / 17.10 | 26.13 | 16.32 |
| | $UMFormer_{25}$ | ResNet18 | 17.32 / 9.48 | 43.44 / 27.75 | 3.15 / 1.60 | 0.06 / 0.03 | 12.76 / 6.82 | 15.35 | 9.13 |
| | $SegDINO_{25}$ | DINOv3-L | 16.57 / 9.03 | 44.81 / 28.87 | 0.55 / 0.28 | 1.26 / 0.63 | 33.41 / 20.05 | 19.32 | 11.77 |
| | **Ours** | DINOv3-L | 64.13 / 47.20 | 55.14 / 38.07 | 2.39 / 1.21 | 0.66 / 0.33 | **82.47 / 70.17** | 40.96 | 31.40 |
| | **Ours** (LoRA) | | **69.43 / 53.18** | **62.32 / 45.27** | **12.88 / 6.88** | **4.30 / 2.20** | 71.38 / 55.49 | **44.06** | **32.60** |
| Multi | $ESANet_{21}$ | ResNet34 | 55.97 / 38.86 | 13.10 / 7.01 | 3.63 / 1.85 | 0.00 / 0.00 | 60.15 / 43.01 | 26.57 | 18.14 |
| | $CMGFNet_{22}$ | ResNet34 | 24.76 / 14.13 | 44.69 / 28.78 | 10.32 / 5.44 | 0.11 / 0.05 | 7.19 / 3.73 | 17.41 | 10.43 |
| | $FTransUNet_{24}$ | R50-ViT-B | 61.17 / 44.06 | 41.71 / 26.35 | 4.73 / 2.42 | 0.00 / 0.00 | 65.28 / 48.45 | 34.58 | 24.26 |
| | $MultiSenseSeg_{24}$ | Segformer-B2 | 49.02 / 32.46 | 43.91 / 28.13 | 3.80 / 1.94 | 0.08 / 0.04 | 48.68 / 32.17 | 29.10 | 18.95 |
| | $MFNet_{25}$ | SAM-L | **68.22 / 51.77** | 57.49 / 40.34 | 8.14 / 4.24 | 6.02 / 3.10 | **85.99 / 75.42** | 45.17 | 34.97 |
| | **Ours** | DINOv3-L | 65.55 / 48.75 | 54.69 / 37.63 | 1.74 / 0.88 | 0.50 / 0.25 | 79.08 / 65.39 | 40.31 | 30.58 |
| | **Ours** (LoRA) | | 66.16 / 49.43 | **64.75 / 47.88** | **17.92 / 9.84** | **23.97 / 13.62** | 73.61 / 58.24 | **49.28** | **35.80** |

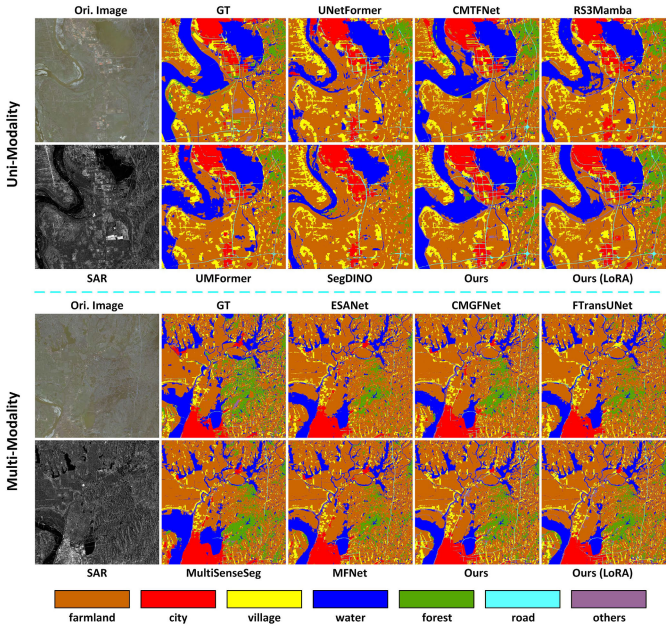



Fig. 7. Visualized comparisons on the WHU-OPT-SAR test set. Specifically, the upper part of the figure presents the results of uni-modal methods, while the lower part shows those of multimodal methods.

effectively reflects whether the learned features are scene-invariant. Results are shown in Table IV.

Analyzing Table IV reveals a significant conclusion: MM-DINO demonstrates far superior generalization performance compared to other methods, in both uni-modal and multimodal settings. We conduct a focused analysis on the results of our own model variants. For the multimodal setup, the Ours (LoRA) version achieves an mIoU of 35.80%, representing a substantial improvement over the Ours version (30.58%). This gain is primarily attributed to significant improvements in the "low vegetation" and "tree" categories, where IoU scores increase by approximately 9.0% and 13.4%, respectively. While the performance on the "car" category decreases slightly (from 65.39% to 58.24% IoU), the overall balance and robustness across all land-cover classes are greatly enhanced. This suggests that LoRA fine-tuning helps the model adapt its feature representation to be more consistent and less biased toward specific, easily transferable classes (like cars), thereby improving generalization on harder, more domain-specific classes. The strong generalization, especially of the Ours baseline, can be attributed to: 1) the frozen DINOv3 backbone, which provides powerful, general-purpose object representations pretrained on natural images, offering strong domain-invariant geometric and semantic understanding; and 2) our multimodal fusion mechanism stably combines RGB texture and DSM height information, providing robust features for objects with distinct 3-D structures and reducing interference from scene-specific color and lighting variations.

TABLE V

ABLATION STUDY OF THE PROPOSED COMPONENTS ON THE VAIHINGEN DATASET. "FRM" DENOTES FEATURE REINFORCEMENT MODULE, "ACFM" DENOTES ADAPTIVE CHANNEL FUSION MODULE, AND "FRN" DENOTES FEATURE REFINEMENT NECK

| Models | Adapter | FRM | ACFM | FRN | Params. | mF1 | mIoU |
|---|---|---|---|---|---|---|---|
| Baseline | × | × | × | × | 0.01 | 80.44 | 69.22 |
| v1 | ✓ | × | × | × | 14.43 | 82.47 | 71.68 |
| v2 | × | ✓ | × | × | 8.97 | 88.99 | 80.65 |
| v3 | × | × | ✓ | × | 2.11 | 89.06 | 80.72 |
| v4 | × | × | × | ✓ | 160.45 | 88.07 | 79.25 |
| v5 | ✓ | ✓ | × | × | 21.10 | 90.36 | 82.82 |
| v6 | ✓ | ✓ | ✓ | × | 21.17 | 90.52 | 83.04 |
| Ours | ✓ | ✓ | ✓ | ✓ | 31.20 | **90.84** | **83.60** |

## E. Ablation Study

To systematically evaluate the contribution of each core module and their synergistic effects across different modalities, we conducted more detailed ablation studies on the Vaihingen dataset under the multimodal input setting. As shown in Table V, we start from a baseline consisting of a frozen backbone with only a segmentation head (baseline), and progressively introduce the Adapter, FRM, ACFM, and the decoder containing the FRN.

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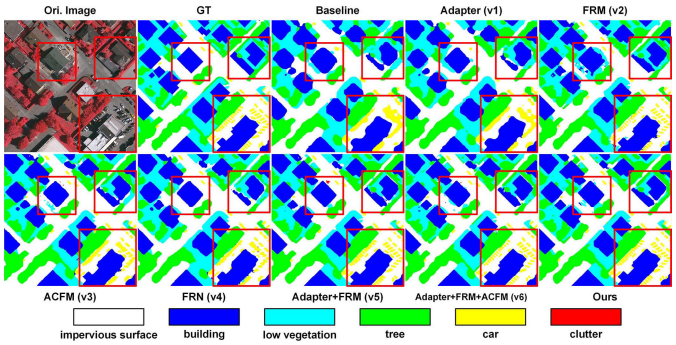



Fig. 8. Visual comparison of ablation studies on the Vaihingen test set. To highlight the differences, some red boxes are added to all subfigures.

The following key conclusions can be drawn from Table V.

*1) Individual Module Contribution:* Introducing the adapter alone (v1) improves performance (mIoU increases from 69.22% to 71.68%), indicating that reshaping transformer sequential features into spatial pyramid features is effective. Introducing FRM alone (v2) or ACFM alone (v3) both significantly enhance performance, achieving comparable results (80.65% and 80.72% mIoU, respectively). This demonstrates that both cross-scale feature enhancement and adaptive channel fusion are effective optimization strategies. Introducing the decoder containing FRN alone (v4) also yields a considerable gain (79.25% mIoU), validating the effectiveness of progressive feature refinement.

*2) Synergistic Effects Between Modules:* Combining the Adapter with FRM (v5) leads to a performance (82.82% mIoU) that significantly exceeds the sum of their individual contributions. This suggests that the solid feature foundation provided by the Adapter unlocks greater potential for the enhancement capabilities of FRM. Further introducing ACFM (v6) achieves a performance of 83.04% mIoU, proving the complementarity between multimodal fusion and feature enhancement.

*3) Advantage of the Complete Architecture:* When the Adapter, FRM, ACFM, and FRN work together (Ours), the model achieves its best performance (83.60% mIoU). This fully validates the rationality of our progressive "decoupling-enhancement-fusion-refinement" design philosophy, where the modules achieve a "$1+1+1+1>4$" effect within the overall framework.

*4) Visual Analysis:* We present representative segmentation results from the ablation study in Fig. 8 to provide an intuitive comparison of each module's effect. It can be observed that the baseline model produces results with blurred edges and severe interclass adhesion, struggling to distinguish fine structures. After introducing FRM (v2), the model shows significant improvement in the geometric integrity of large-scale objects like buildings and roads, though the results still contain considerable fragmented patches. By further introducing the Adapter and ACFM on top of v2 (v6), this fragmentation is noticeably alleviated, and object boundaries become more coherent. Finally, the complete model (Ours) achieves the best balance between geometric completeness and boundary precision, maintaining clear boundaries while producing more compact object interiors, demonstrating the effectiveness of the progressive feature refinement mechanism.

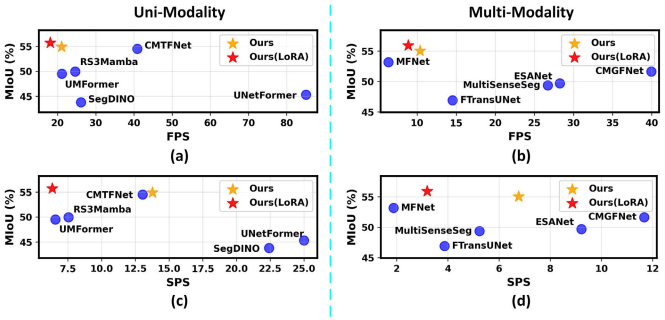



Fig. 9. Tradeoff between inference/training speed and segmentation accuracy on the WHU-OPT-SAR test set (on $256\times256$ resolution images, batch size = 1). (a) Inference speed (FPS) versus mIoU on uni-modal tasks. (b) Inference speed (FPS) versus mIoU on multimodal tasks. (c) Training throughput (SPS) versus mIoU on uni-modal tasks. (d) Training throughput (SPS) versus mIoU on multimodal tasks. "FPS": Frames Per Second. "SPS": Steps Per Second.

TABLE VI

COMPLEXITY AND EFFICIENCY COMPARISON ON THE WHU-OPT-SAR TEST SET (ON 256 × 256 RESOLUTION IMAGES, BATCH SIZE = 1). "PARAMS.": TRAINABLE PARAMETERS (TOTAL PARAMETERS IN BRACKETS). "MEMORY": GPU MEMORY USAGE. S DENOTES USING DINOV3-SMALL AS THE FROZEN BACKBONE, L DENOTES DINOV3-LARGE, AND L-LORA DENOTES DINOV3-LARGE WITH LORA FINE-TUNING

| Modality | Models | | Params. (M) | FLOPs (G) | Memory (MB) | mIoU |
|---|---|---|---|---|---|---|
| Uni | $\text{UNetFormer}_{22}$ | | 11.69 | 2.94 | 65.03 | 45.30 |
| | $\text{CMTFNet}_{23}$ | | 30.07 | 8.57 | 180.68 | 54.50 |
| | $\text{RS3Mamba}_{24}$ | | 43.32 | 15.83 | 253.33 | 49.95 |
| | $\text{UMFormer}_{25}$ | | 12.37 | 2.98 | 81.69 | 49.51 |
| | $\text{SegDINO}_{25}$ | | 3.14 (306.29) | 60.17 | 1195.00 | 43.77 |
| | **Ours** | S | 29.33 (50.93) | 39.02 | 281.28 | 54.12 |
| | | L | 31.13 (334.29) | 94.60 | 1351.58 | 54.92 |
| | | L-LoRA | 31.43 (334.58) | 94.80 | 1352.71 | **55.72** |
| Multi | $\text{ESANet}_{21}$ | | 46.89 | 9.49 | 202.50 | 49.69 |
| | $\text{CMGFNet}_{22}$ | | 43.68 | 38.84 | 237.94 | 51.65 |
| | $\text{FTransUNet}_{24}$ | | 203.60 | 56.11 | 826.59 | 46.92 |
| | $\text{MultiSenseSeg}_{24}$ | | 58.44 | 66.11 | 390.63 | 49.38 |
| | $\text{MFNet}_{25}$ | | 8.98 (317.25) | 147.72 | 1307.14 | 53.18 |
| | **Ours** | S | 29.40 (51.00) | 54.63 | 301.69 | 54.12 |
| | | L | 31.20 (334.36) | 165.80 | 1374.80 | 55.04 |
| | | L-LoRA | 31.49 (334.65) | 165.95 | 1375.92 | **55.92** |

## F. Efficiency Analysis

In practical applications, model efficiency is as important as accuracy. Under the setting of $256\times256$ resolution and a batch size of 1, we compare MM-DINO with other advanced models in terms of parameter count, floating-point operations (FLOPs), GPU memory consumption, inference speed (FPS), and training throughput (SPS). Results are shown in Table VI and Fig. 9.

It is essential to view the efficiency data objectively: since MM-DINO employs the large-scale DINOv3-L as its frozen backbone, its parameter count (~31M trainable, 334M total) and FLOPs are indeed higher compared to some compact CNN networks (e.g., UNetFormer). However, the core of efficiency analysis lies in the accuracy-efficiency tradeoff. Notably, our framework offers flexible configurations by utilizing different scales of the DINOv3 backbone. The S variant (DINOv3-Small) provides a more lightweight option with significantly reduced computational cost while still achieving

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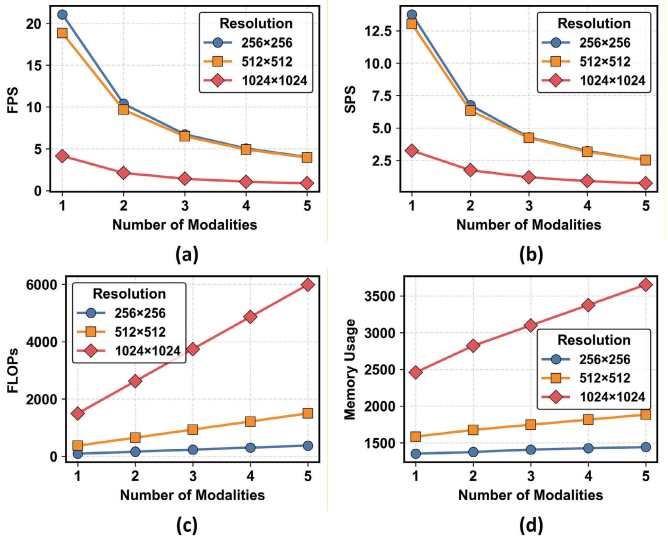



Fig. 10. Efficiency analysis of MM-DINO with DINOv3-L backbone under different input resolutions and varying numbers of modalities (batch size = 1). (a) Inference speed (FPS) versus number of modalities. (b) Training throughput (SPS) versus number of modalities. (c) FLOPs versus number of modalities. (d) GPU memory usage versus number of modalities. All metrics are measured on a single NVIDIA L20 GPU with 48-GB memory. The three curves in each subfigure correspond to input resolutions of 256 × 256, 512 × 512, and 1024 × 1024 pixels, respectively.

competitive accuracy, whereas the L and L-LoRA variants pursue higher performance at the cost of increased resources.

*1) Comparison with Models of Similar Accuracy:* In the uni-modal group, the S variant attains 54.12% mIoU with only 39.02-G FLOPs and 29.33-M trainable parameters (50.93-M total), which is on par with CMTFNet (54.50% mIoU) but with moderately higher FLOPs. Meanwhile, the L-LoRA variant pushes the mIoU to 55.72% at the expense of 94.80-G FLOPs. This demonstrates that even the lightweight S configuration can match or exceed the accuracy of many existing methods while maintaining a favorable efficiency profile. In the multimodal setting, the S model achieves 54.12% mIoU with 54.63-G FLOPs, whereas the top-performing L-LoRA reaches 55.92% mIoU with 165.95-G FLOPs—a 1.8 percentage point gain for roughly three times the computational cost. This tradeoff underscores the adaptability of MM-DINO to different resource constraints.

*2) Comparison with Models Using Similar-Scale Backbones:* Compared to compact networks like UNetFormer (2.94-G FLOPs, 45.30% mIoU), the S variant delivers a substantial accuracy improvement (54.12% mIoU) with still moderate FLOPs, highlighting the benefit of leveraging a frozen foundation model even at a smaller scale. When compared to methods that also employ large foundation models, such as MFNet (SAM-L-based), our L-LoRA variant achieves higher mIoU on multimodal WHU-OPT-SAR (55.92% versus 53.18%) with comparable trainable parameters, while also demonstrating significantly better FPS (8.88 versus 6.33) and SPS (3.18 versus 1.88), confirming its superior inference and training efficiency.

*3) Scalability With Resolution and Number of Modalities:* To systematically evaluate the computational behavior of MM-DINO under varying conditions, we analyze its performance on the DINOv3-L backbone across three resolutions (256 × 256, 512 × 512, 1024 × 1024) and up to five input modalities. As shown in Fig. 10, both FLOPs and GPU memory grow almost linearly with the number of modalities, while the trainable parameters increase only marginally (from 31.13 to 31.30 M) due to the lightweight nature of the adaptive fusion weights. The cost difference between 256 × 256 and 512 × 512 is relatively modest, but scaling to 1024 × 1024 causes a sharp rise in FLOPs and memory, accompanied by a steep drop in FPS and SPS—likely reflecting the computational and memory bandwidth limits of the L20 GPU. The FPS and SPS curves exhibit a concave shape: the largest drop occurs when moving from one to two modalities, indicating that the initial overhead of enabling multimodal processing dominates, while further additions incur progressively smaller penalties. These results demonstrate that MM-DINO offers predictable scalability and can be flexibly adapted to different resource constraints by choosing an appropriate resolution and backbone scale.

### G. Discussion on Limitations and Future Directions

Despite its strong performance, MM-DINO has several limitations that warrant future investigation.

First, the framework inherits the input constraints of its frozen DINOv3 backbone, which is designed for three-channel RGB inputs. This inherently limits direct applicability to multispectral or hyperspectral imagery with more bands. Moreover, when integrating single-channel modalities (e.g., DSM and SAR), they must be replicated to three channels to match the backbone's input format, introducing computational redundancy without fully leveraging the unique characteristics of each modality. A promising direction is to adapt the backbone itself: recent work on dual-teacher distillation [47] demonstrates that the representational power of optical foundation models can be transferred to multispectral data, suggesting a pathway to extend MM-DINO to a wider spectrum of remote sensing modalities.

Second, while our adaptive fusion mechanism is lightweight and flexible, its current design performs early interaction primarily at the feature level. For highly heterogeneous modalities such as optical and SAR, a more principled fusion strategy that accounts for their distinct physical acquisition principles could be beneficial. Incorporating ideas from physics-guided deep neural networks [48] could embed physical constraints directly into the fusion process, improving both robustness and interpretability when integrating data from fundamentally different sensors.

Third, the robustness of MM-DINO under common data degradations—such as noise, atmospheric interference, or spectral variability—has not been systematically evaluated. Future work should investigate this aspect, potentially drawing insights from fields like hyperspectral anomaly detection. For instance, self-attention-based suppression mechanisms [49] could be adapted to enhance the model's resilience to outliers and corrupted inputs.

Addressing these limitations would not only strengthen MM-DINO's performance but also broaden its applicability

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to the diverse and often challenging data encountered in real-world Earth observation tasks.

## V. Conclusion

This article presented MM-DINO, a universal and efficient framework for uni-modal and multimodal remote sensing semantic segmentation, built upon a frozen DINOv3 backbone. By introducing a modality-adaptive adapter and a feature enhancement and refinement decoder, we effectively bridge the gap between general-purpose vision foundation models and the specific demands of dense prediction in remote sensing. The adapter transforms sequential features into spatial pyramids and enables early, soft cross-modal interaction, while the decoder progressively enhances, fuses, and refines these features to produce high-quality segmentation maps.

Extensive experiments on the ISPRS Vaihingen, Potsdam, and WHU-OPT-SAR datasets validate the effectiveness of our design. MM-DINO achieves state-of-the-art performance in both uni-modal and multimodal settings. More importantly, it demonstrates exceptional cross-dataset generalization capability, significantly outperforming existing methods in zero-shot transfer scenarios. This robustness is attributed to the preservation of the foundation model's general knowledge, combined with our lightweight, task-specific adaptation. Furthermore, efficiency analysis confirms that MM-DINO strikes a favorable balance between accuracy and computational cost, offering flexible configurations to suit different resource constraints.

Despite these advances, the framework has several limitations, including its reliance on a three-channel backbone, the need for more principled fusion of highly heterogeneous modalities, and unexplored robustness to common data degradations. Future work will focus on: 1) extending the backbone to natively support multispectral inputs via cross-modal distillation; 2) incorporating physics-guided constraints into the fusion process for more interpretable cross-modal learning; and 3) enhancing model resilience through suppression mechanisms inspired by anomaly detection. We believe MM-DINO provides a strong and scalable baseline, paving the way for broader adoption of publicly available foundation models in complex Earth observation tasks.

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